PARTIAL DERIVATIVE

What is Partial Derivative

- The partial derivative is used in vector calculus and differential geometry. In Mathematics, sometimes the function depends on two or more variables. Here, the derivative converts into the partial derivative since the function depends on several variables.

Used instead of "d" in usual $\frac{df}{dx}$ notation to emphasize that this is a partial derivative.

$$\frac{\overbrace{\partial f}^{\text{Can be thought of as "a tiny change in the function's output"}}}{\underbrace{\partial x}_{\text{Can be thought of as "a tiny change in } x \text{"}}}$$

Multivariable function

Indicates which input variable is changed slightly.

EXAMPLES

Example 1: Determine the partial derivative of the function: $f(x,y) = 3x + 4y$.

Solution:

Given function: $f(x,y) = 3x + 4y$

To find $\partial f/\partial x$, keep y as constant and differentiate the function:

Therefore, $\partial f/\partial x = 3$

Similarly, to find $\partial f/\partial y$, keep x as constant and differentiate the function:

Therefore, $\partial f/\partial y = 4$

Example 2: Find the partial derivative of $f(x,y) = x^2y + \sin x + \cos y$.

Solution:

Now, find out f_x first keeping y as constant

$f_x = \partial f/\partial x = (2x)\, y + \cos x + 0$

$= 2xy + \cos x$

When we keep y as constant cos y becomes a constant so its derivative becomes zero.

Similarly, finding f_y

$f_y = \partial f/\partial y = x^2 + 0 + (-\sin y)$

$= x^2 - \sin y$

▸ **Example 1** Find $f_x(1, 3)$ and $f_y(1, 3)$ for the function $f(x, y) = 2x^3y^2 + 2y + 4x$.

Solution. Since

$$f_x(x, 3) = \frac{d}{dx}[f(x, 3)] = \frac{d}{dx}[18x^3 + 4x + 6] = 54x^2 + 4$$

we have $f_x(1, 3) = 54 + 4 = 58$. Also, since

$$f_y(1, y) = \frac{d}{dy}[f(1, y)] = \frac{d}{dy}[2y^2 + 2y + 4] = 4y + 2$$

we have $f_y(1, 3) = 4(3) + 2 = 14$. ◂

Example 3 Find $\partial z/\partial x$ and $\partial z/\partial y$ if $z = x^4 \sin(xy^3)$.

Solution.

$$\begin{aligned}
\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial x}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial x}(x^4) \\
&= x^4 \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3 = x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3) \\
\frac{\partial z}{\partial y} &= \frac{\partial}{\partial y}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial y}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial y}(x^4) \\
&= x^4 \cos(xy^3) \cdot 3xy^2 + \sin(xy^3) \cdot 0 = 3x^5 y^2 \cos(xy^3)
\end{aligned}$$

▸ **Example 4** Recall that the wind chill temperature index is given by the formula

$$W = 35.74 + 0.6215T + (0.4275T - 35.75)v^{0.16}$$

Compute the partial derivative of W with respect to v at the point $(T, v) = (25, 10)$ and interpret this partial derivative as a rate of change.

Solution. Holding T fixed and differentiating with respect to v yields

$$\frac{\partial W}{\partial v}(T, v) = 0 + 0 + (0.4275T - 35.75)(0.16)v^{0.16-1} = (0.4275T - 35.75)(0.16)v^{-0.84}$$

Since W is in degrees Fahrenheit and v is in miles per hour, a rate of change of W with respect to v will have units °F/(mi/h) (which may also be written as °F·h/mi). Substituting

$T = 25$ and $v = 10$ gives

$$\frac{\partial W}{\partial v}(25, 10) = (-4.01)10^{-0.84} \approx -0.58 \ \frac{°\text{F}}{\text{mi/h}}$$

as the instantaneous rate of change of W with respect to v at $(T, v) = (25, 10)$. We conclude that if the air temperature is a constant 25°F and the wind speed changes by a small amount from an initial speed of 10 mi/h, then the ratio of the change in the wind chill index to the change in wind speed should be about −0.58°F/(mi/h). ◂

Example 5 Let $f(x, y) = x^2y + 5y^3$.

(a) Find the slope of the surface $z = f(x, y)$ in the x-direction at the point $(1, -2)$.

(b) Find the slope of the surface $z = f(x, y)$ in the y-direction at the point $(1, -2)$.

Solution (a). Differentiating f with respect to x with y held fixed yields

$$f_x(x, y) = 2xy$$

Thus, the slope in the x-direction is $f_x(1, -2) = -4$; that is, z is decreasing at the rate of 4 units per unit increase in x.

Solution (b). Differentiating f with respect to y with x held fixed yields

$$f_y(x, y) = x^2 + 15y^2$$

Thus, the slope in the y-direction is $f_y(1, -2) = 61$; that is, z is increasing at the rate of 61 units per unit increase in y. ◀

Example 12 Find the second-order partial derivatives of $f(x, y) = x^2y^3 + x^4y$.

olution. We have

$$\frac{\partial f}{\partial x} = 2xy^3 + 4x^3y \quad \text{and} \quad \frac{\partial f}{\partial y} = 3x^2y^2 + x^4$$

o that

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}(2xy^3 + 4x^3y) = 2y^3 + 12x^2y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}(3x^2y^2 + x^4) = 6x^2y$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}(3x^2y^2 + x^4) = 6xy^2 + 4x^3$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}(2xy^3 + 4x^3y) = 6xy^2 + 4x^3 \blacktriangleleft$$

Example 12 Find the second-order partial derivatives of $f(x, y) = x^2y^3 + x^4y$.

Practice Sums

1. $z = e^{(x^2+y^2)}$. find $\partial z/\partial x$

$\therefore \dfrac{\partial z}{\partial x} = e^{x^2+y^2} \times 2x$

$= 2x\left(e^{x^2+y^2}\right)$

2. $z = \sin x \cos xy$

$\dfrac{\partial z}{\partial x} = \{(\sin x) \times (-y\sin(xy))\} + \cos(xy) \times \cos x$

$= -y\sin x \sin(xy) + \cos(xy)\cos x.$

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4. $f(x,y) = xe^{-y} + 5y$

a) $z = f(x,y)$ $(4,0)$

$\therefore \dfrac{\partial z}{\partial x} = e^{-y}$

$= e^{-0}$

$= 1$

b) $\dfrac{\partial z}{\partial y} = -xye^{-y} + 5$

$= -4(0)e^{-0} + 5$

$= 5.$

22. $z = \dfrac{x^3y^4}{\sqrt{x+y}}$

$$\frac{\partial z}{\partial x} = \frac{(x+y)^{1/2}(y^4 \times 3x^2) - \left(x^3y^4 \times \frac{(x+y)^{-1/2}}{2}\right)}{(\sqrt{x+y})^2}$$

$$= \frac{(x+y)^{1/2}(3y^4x^2) - \dfrac{x^3y^4}{2(x+y)^{1/2}}}{(x+y)}$$

$$= \frac{(x+y)(3y^4x^2) - x^3y^4}{2(x+y)^{3/2}}$$

$$= \frac{3x^3y^4 + 3y^5x^2 - x^3y^4}{2(x+y)^{3/2}}$$

$$= \frac{2x^3y^4 + 3y^5x^2}{2(x+y)^{3/2}}$$

$$= \frac{x^2y^4(2x+3y)}{2(x+y)^{3/2}}$$

Q. Practice Problem

$f(x,y,z) = \sqrt{xy} + \ln(x^2 z^3) - x\tan(z)$; find f_{xyz}

$$\frac{\partial f}{\partial x} = \left\{\frac{1}{2} \times (xy)^{-1/2} \times y\right\} + \left\{\frac{1}{x^2 z^3} \times z^3 \times 2x\right\} - \tan(z)$$

$$= \frac{y}{2\sqrt{xy}} + \frac{2}{x} - \tan(z)$$

$$= \frac{x^{-1/2} y^{1/2}}{2} + 2x^{-1} - \tan(z)$$

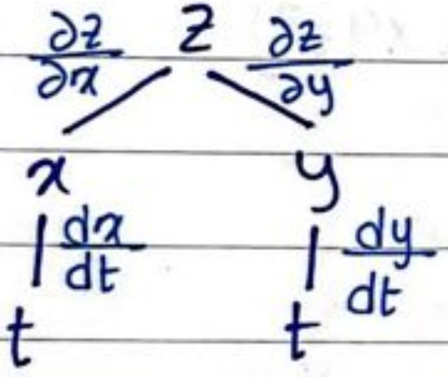
$$f_{xy} = \frac{1}{2} x^{-1/2} \times \frac{1}{2} y^{1/2 - 1} + 0 - 0$$

$$\therefore f_{xyz} = 0$$

$$= \frac{1}{4\sqrt{xy}}$$

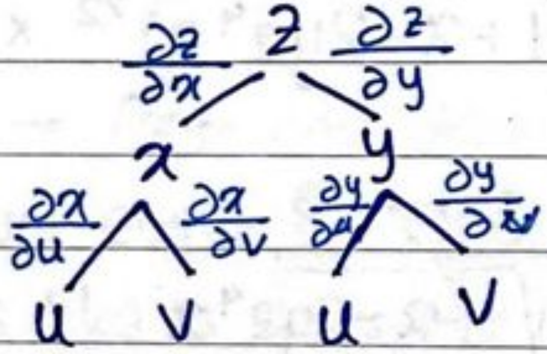
CHAIN RULE

$z \to xy \to t$



z depends on xy and x depends on t, and y depends on t.

$z \to xy \to uv$



z depends on x & y. x depends on u & v. y depends on u and v

** when 2 variables, use ∂. When one variable, use d.

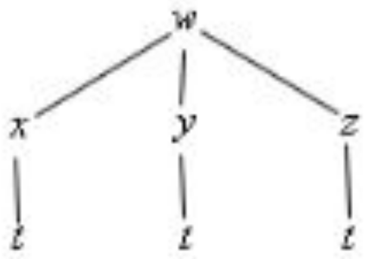
$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \times \frac{dx}{dt} + \frac{\partial z}{\partial y} \times \frac{dy}{dt}$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \times \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \times \frac{\partial y}{\partial u}$$

EXAMPLES

(a) $\dfrac{dw}{dt}$ for $w = f(x, y, z)$, $x = g_1(t)$, $y = g_2(t)$, and $z = g_3(t)$ Hide Solution ▾

So, we'll first need the tree diagram so let's get that.

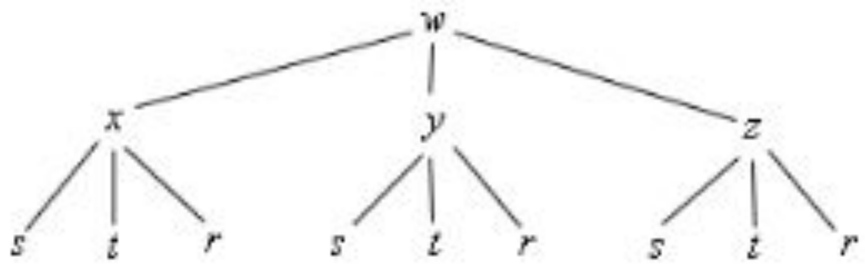


From this it looks like the chain rule for this case should be,

$$\frac{dw}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

(b) $\dfrac{\partial w}{\partial r}$ for $w = f(x, y, z)$, $x = g_1(s, t, r)$, $y = g_2(s, t, r)$, and $z = g_3(s, t, r)$ Hide Solution ▾

Here is the tree diagram for this situation.



From this it looks like the derivative will be,

$$\frac{\partial w}{\partial r} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial r}$$

Example 1 Suppose that

$$z = x^2 y, \quad x = t^2, \quad y = t^3$$

Use the chain rule to find dz/dt, and check the result by expressing z as a function of t and differentiating directly.

Solution. By the chain rule [Formula (5)],

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt} = (2xy)(2t) + (x^2)(3t^2) \\ &= (2t^5)(2t) + (t^4)(3t^2) = 7t^6\end{aligned}$$

Alternatively, we can express z directly as a function of t,

$$z = x^2 y = (t^2)^2(t^3) = t^7$$

and then differentiate to obtain $dz/dt = 7t^6$. However, this procedure may not always be convenient. ◄

Example 2 Suppose that

$$w=\sqrt{x^2+y^2+z^2}, \quad x=\cos\theta, \quad y=\sin\theta, \quad z=\tan\theta$$

Use the chain rule to find $dw/d\theta$ when $\theta=\pi/4$.

Solution. From Formula (6) with θ in the place of t, we obtain

$$\begin{aligned}\frac{dw}{d\theta}&=\frac{\partial w}{\partial x}\frac{dx}{d\theta}+\frac{\partial w}{\partial y}\frac{dy}{d\theta}+\frac{\partial w}{\partial z}\frac{dz}{d\theta}\\&=\frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2x)(-\sin\theta)+\frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2y)(\cos\theta)\\&\quad+\frac{1}{2}(x^2+y^2+z^2)^{-1/2}(2z)(\sec^2\theta)\end{aligned}$$

When $\theta=\pi/4$, we have

$$x=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}, \quad y=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}, \quad z=\tan\frac{\pi}{4}=1$$

Substituting $x=1/\sqrt{2}$, $y=1/\sqrt{2}$, $z=1$, $\theta=\pi/4$ in the formula for $dw/d\theta$ yields

$$\begin{aligned}\left.\frac{dw}{d\theta}\right|_{\theta=\pi/4}&=\frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)(\sqrt{2})\left(-\frac{1}{\sqrt{2}}\right)+\frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)(\sqrt{2})\left(\frac{1}{\sqrt{2}}\right)+\frac{1}{2}\left(\frac{1}{\sqrt{2}}\right)(2)(2)\\&=\sqrt{2}\end{aligned}$$

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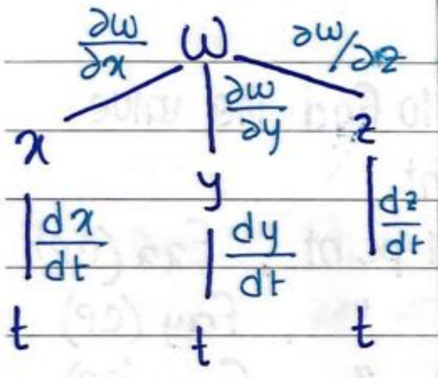
10. $\omega = \sqrt{1+x-2yz^4x}$; $x = \ln t$, $y = t$, $z = 3t$ (find $d\omega/dt$)



$$\frac{d\omega}{dt} = \frac{\partial\omega}{\partial x}\times\frac{dx}{dt} + \frac{\partial\omega}{\partial y}\times\frac{dy}{dt} + \frac{\partial\omega}{\partial z}\times\frac{dz}{dt}$$

$$\frac{\partial\omega}{\partial x} = \frac{1}{2}\times(1+x-2yz^4x)^{-1/2}\times(0+1-2yz^4)$$

$$\frac{dx}{dt} = \frac{1}{t}$$

$$\frac{\partial\omega}{\partial y} = \frac{1}{2}(1+x-2yz^4x)^{-1/2}\times(0+0-2z^4x)$$

$$\frac{\partial\omega}{\partial y} = \frac{-2z^4x}{2\sqrt{1+x-2yz^4x}}$$

$$\frac{dy}{dt} = 1$$

$$\frac{\partial\omega}{\partial z} = \frac{1}{2}(1+x-2yz^4x)^{-1/2}\times(0+0-2yx\times 4z^3)$$

$$= \frac{-8yxz^3}{2\sqrt{1+x-2yz^4x}}$$

$$\frac{dz}{dt} = 3$$

$$\therefore \frac{d\omega}{dt} = \frac{(1-2yz^4)\times\frac{1}{t} - (2z^4x\times 1) - (8yxz^3\times 3)}{2\sqrt{1+x-2yz^4x}}$$

** put $x = \ln t$, $y = t$ and $z = 3t$ and simplify.

(a) $z = x\mathbf{e}^{xy}$, $x = t^2$, $y = t^{-1}$ Hide Solution ▾

There really isn't all that much to do here other than using the formula.

$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} \\ &= \left(\mathbf{e}^{xy} + yx\mathbf{e}^{xy}\right)(2t) + x^2\mathbf{e}^{xy}\left(-t^{-2}\right) \\ &= 2t\left(\mathbf{e}^{xy} + yx\mathbf{e}^{xy}\right) - t^{-2}x^2\mathbf{e}^{xy}\end{aligned}$$

So, technically we've computed the derivative. However, we should probably go ahead and substitute in for x and y as well at this point since we've already got t's in the derivative. Doing this gives,

$$\frac{dz}{dt} = 2t\left(\mathbf{e}^{t} + t\mathbf{e}^{t}\right) - t^{-2}t^4\mathbf{e}^{t} = 2t\mathbf{e}^{t} + t^2\mathbf{e}^{t}$$

Note that in this case it might actually have been easier to just substitute in for x and y in the original function and just compute the derivative as we normally would. For comparison's sake let's do that.

$$z = t^2\mathbf{e}^{t} \qquad \Rightarrow \qquad \frac{dz}{dt} = 2t\mathbf{e}^{t} + t^2\mathbf{e}^{t}$$

Example 2 $\frac{dz}{dx}$ for $z = x\ln(xy) + y^3$, $y = \cos(x^2+1)$

Hide Solution ▾

We'll just plug into the formula.

$$
\begin{aligned}
\frac{dz}{dx} &= \left(\ln(xy) + x\frac{y}{xy}\right) + \left(x\frac{x}{xy} + 3y^2\right)\left(-2x\sin(x^2+1)\right) \\
&= \ln(x\cos(x^2+1)) + 1 - 2x\sin(x^2+1)\left(\frac{x}{\cos(x^2+1)} + 3\cos^2(x^2+1)\right) \\
&= \ln(x\cos(x^2+1)) + 1 - 2x^2\tan(x^2+1) - 6x\sin(x^2+1)\cos^2(x^2+1)
\end{aligned}
$$

Example 3 Find $\dfrac{\partial z}{\partial s}$ and $\dfrac{\partial z}{\partial t}$ for $z = \mathbf{e}^{2r}\sin(3\theta)$, $r = st - t^2$, $\theta = \sqrt{s^2 + t^2}$.

Hide Solution ▾

Here is the chain rule for $\dfrac{\partial z}{\partial s}$.

$$\begin{aligned}\frac{\partial z}{\partial s} &= \left(2\mathbf{e}^{2r}\sin(3\theta)\right)(t) + \left(3\mathbf{e}^{2r}\cos(3\theta)\right)\frac{s}{\sqrt{s^2+t^2}} \\ &= t\left(2\mathbf{e}^{2(st-t^2)}\sin\left(3\sqrt{s^2+t^2}\right)\right) + \frac{3s\mathbf{e}^{2(st-t^2)}\cos\left(3\sqrt{s^2+t^2}\right)}{\sqrt{s^2+t^2}}\end{aligned}$$

Now the chain rule for $\dfrac{\partial z}{\partial t}$.

$$\begin{aligned}\frac{\partial z}{\partial t} &= \left(2\mathbf{e}^{2r}\sin(3\theta)\right)(s-2t) + \left(3\mathbf{e}^{2r}\cos(3\theta)\right)\frac{t}{\sqrt{s^2+t^2}} \\ &= (s-2t)\left(2\mathbf{e}^{2(st-t^2)}\sin\left(3\sqrt{s^2+t^2}\right)\right) + \frac{3t\mathbf{e}^{2(st-t^2)}\cos\left(3\sqrt{s^2+t^2}\right)}{\sqrt{s^2+t^2}}\end{aligned}$$